

FEET202

CHINES & TRANSFORMERS

MODULE: 5

ecture:2 – Vector Groups

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Syllabus

g of copper –rating of autotransformers.

er – construction- difference between power transformer and
–Different connections of 3-phase transformers. Y-Y, Δ - Δ , Y- Δ ,
plings – Yy0, Dd0, Yd1, Yd11, Dy1, Dy11. Parallel operation of
s.

ner – stabilization by tertiary winding. Tap changing transformers
on load tap changing, dry type transformers.

Vector Groups

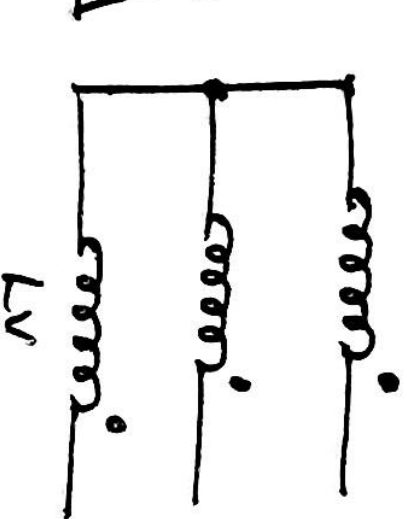
transformers, there can be phase difference and secondary voltages.

ieved by making slight changes in the winding indicates the phase difference between primary and es and thus the winding configuration.

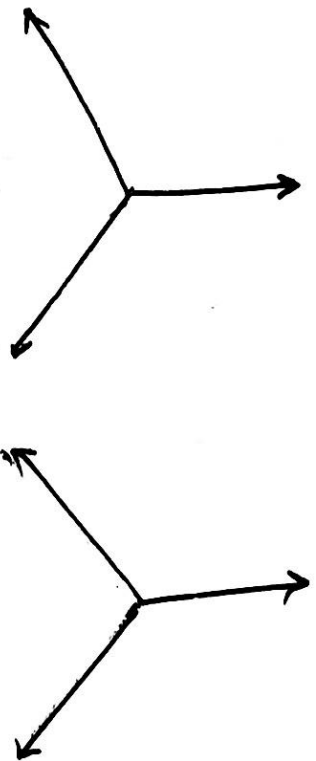
roups are Yy0, Yy6, Dd0, Dd6, Yd1, Yd11, Dy1 and oups, first letter indicates the type of connection and letter indicates the type of connection of LV indicates the phase shift.

Yy0

LV side are star connected.
shift between voltages.



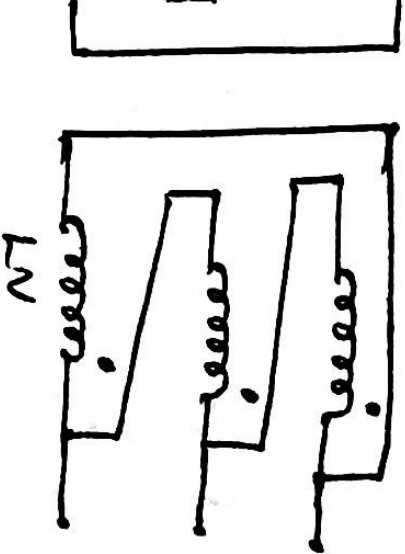
Connection Diagram



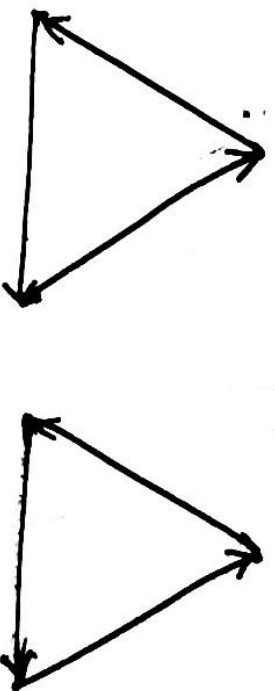
Phasor Diagram

Dd0

LV side are delta connected.
shift between voltages.



Connection Diagram

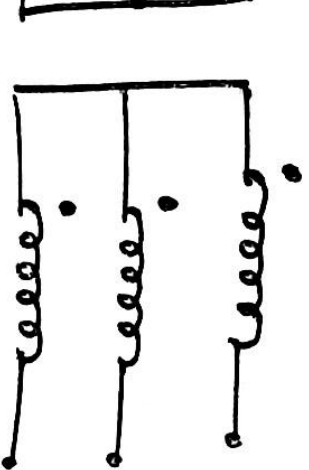


Phasor Diagram

Yy6

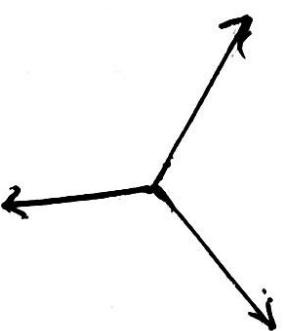
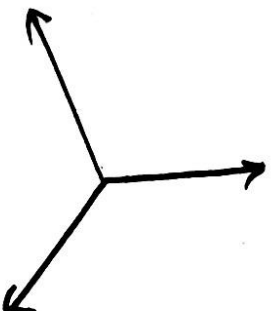
LV side are star connected.

shift of 180° between primary and secondary



LV

ction Diagram

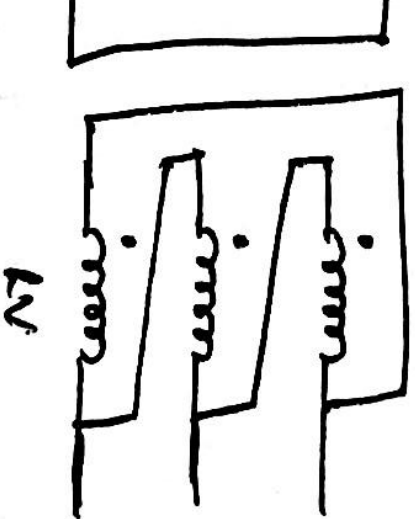


Phasor Diagram

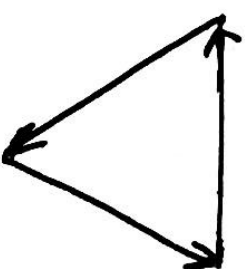
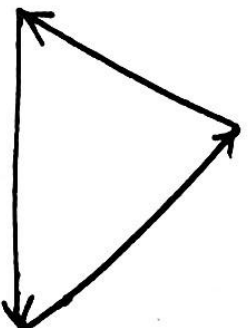
Dd6

LV side are delta connected.

shift of 180° between primary and secondary



Connection Diagram

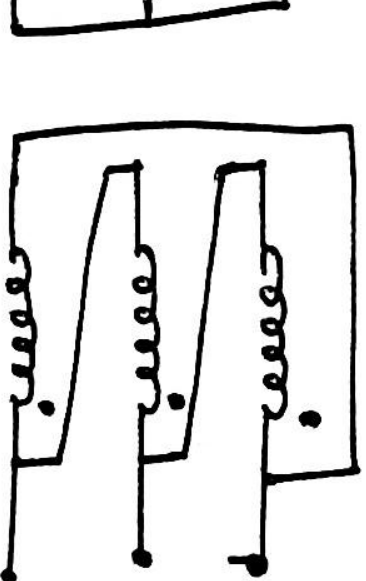


Phasor Diagram

Yd1

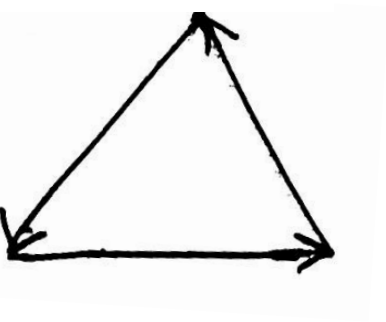
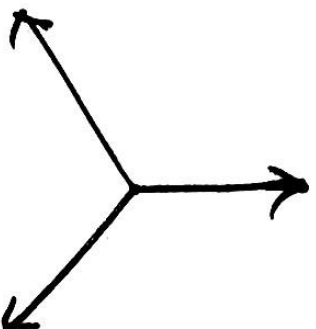
connected whereas LV side is delta connected.

lags the voltage at HV side by 30° .



LV

ction Diagram

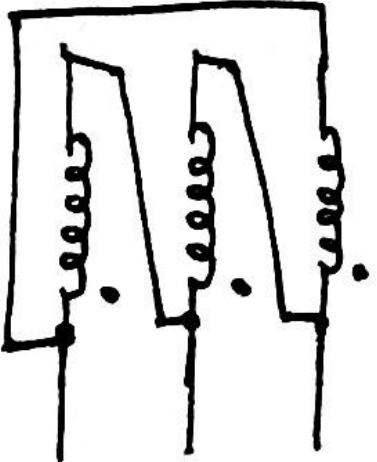


Phasor Diagram

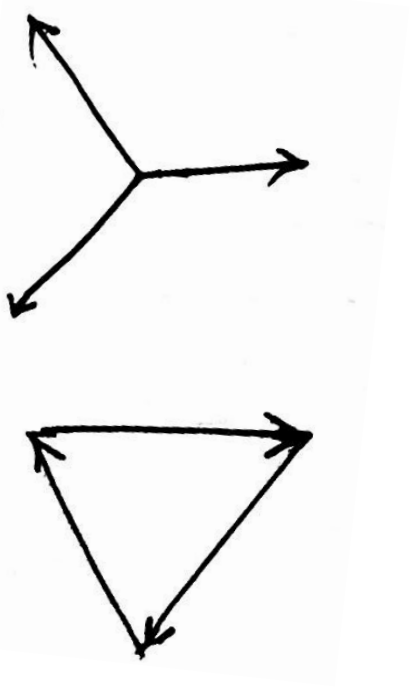
Yd11

connected whereas LV side is delta connected.

leads the voltage at HV side by 30° .



Diagram

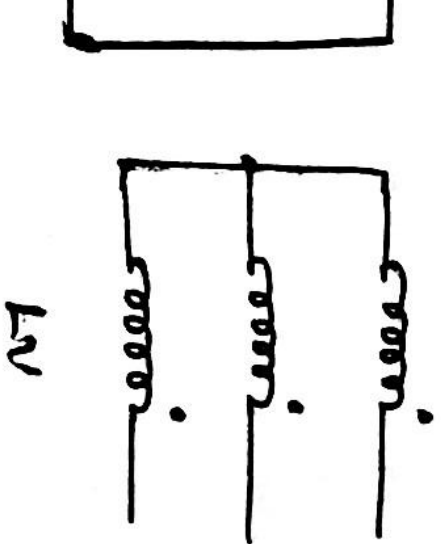


Phasor Diagram

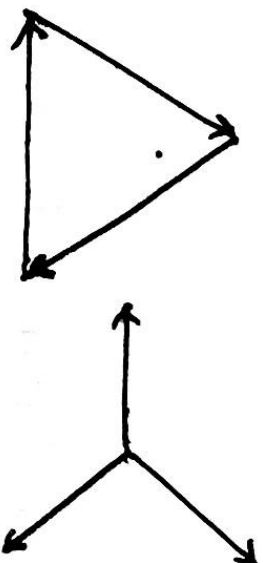
Dy1

connected whereas LV side is star connected.

lags the voltage at HV side by 30° .



Connection Diagram

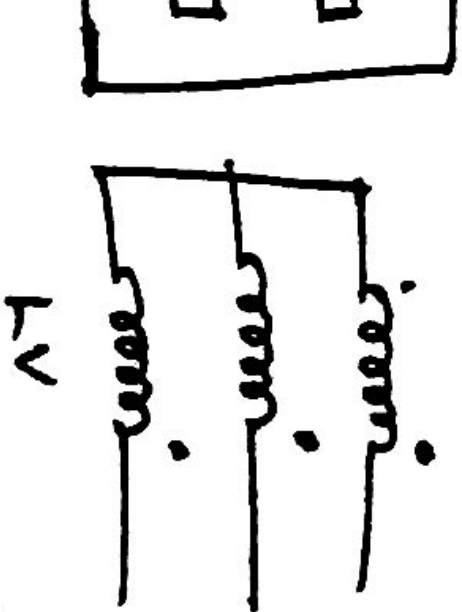


Phasor Diagram

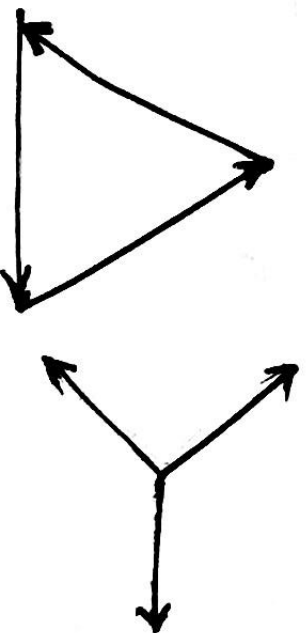
Dy11

connected whereas LV side is star connected.

leads the voltage at HV side by 30° .



Connection Diagram



Phasor Diagram